

Course title: Digital Logic Design

Course code: EENG2041

Credit hours: 3 ECTS: 5 Contact hrs: 2 Lab hrs: 3 Tutorial hrs: 0

Prerequisite: None

Course category: Supportive

Year: II

Semester: I

Department: Computer science

Instructor's name; Yeabsra Dagnaw

Course outline

- 1. Introduction to Digital Systems (2 lecture hour)**
 - 1.1 Digital and analogue quantities
 - 1.2 Binary digit logic level and digital waveform
- 2. Number system, operations and codes (2 lecture hour)**
 - 2.1 Decimal number
 - 2.2 Binary number
 - 2.3 Decimal to binary conversation
 - 2.4 1's and 2's compliment of binary number
 - 2.5 Signed number
 - 2.6 Hexadecimal number
 - 2.7 Octal number
 - 2.8 BCD
- 3. Logic gates (2 lecture hour)**
 - 3.1 The inverter
 - 3.2 The AND gate
 - 3.3 The OR gate
 - 3.4 The NAND gate
 - 3.5 The NOR gate
 - 3.6 The Exclusive OR and Exclusive NOR gates
- 4. Boolean algebra and Logic simplification (4 lecture hour)**
 - 4.1 Boolean operation and expression
 - 4.2 Laws and rules of Boolean algebra
 - 4.3 De Morgan Theorems
 - 4.4 Boolean analysis of logic circuit
 - 4.5 The K-map
- 5. Combinational logic (6 lecture hour)**
 - 5.1 Functions of combinational logic
 - 5.2 Basic combinational logic circuits
 - 5.3 Implementing Combinational logic
 - 5.4 Universal property of NAND and NOR gates
 - 5.5 Adders, decoders, encoders, multiplexers and de multiplexers
- 6. Flip flops (4 lecture hour)**

- 1.1 Latches
- 1.2 Edge triggered flip flops
- 1.3 Master slave flip flops
- 1.4 Applications
- 7. Counters (4 lecture hour)**
 - 7.1 Synchronous counters
 - 7.2 Asynchronous counters
 - 7.3 Up/down counters
 - 7.4 Design of synchronous counters
- 8. Shift registers (4 lecture hour)**
 - 8.1 Basic shift registers
 - 8.2 Serial in serial out registers
 - 8.3 Serial in parallel out Registers
- 9. Memory and storage (4 lecture hour)**
 - 9.1 Basics of semiconductor Memory
 - 9.2 Random access memory (RAM's)
 - 9.3 Read only memory (ROM's)
 - 9.4 Programmable ROMs (PROM, EEPROM)
 - 9.5 Flash memories

Assessment Methods

Assignment/quizzes	10 %
Project	20%
Mid semester examination	20%
Final examination	50%

Text books and References:

1. Morris M. Mano: Digital Design (3rd Edition)
2. R. J. Tocci and N. S. Widmer: Digital Systems – Principles and Applications, 9th Ed, Prentice Hall, 2004
3. T.L. Floyd: Digital Fundamentals, 9th edition ,Prentice Hall
4. Stephen Brown, Zvonko Vranesic: Fundamentals of Digital Logic with Verilog Design, McGraw-Hill Science/Engineering/Math; 1st edition 2002
5. R.P. Jain: Modern Digital Electronics, raw-Hill Science/Engineering/Math; 1 edition, (August 21, 2006)
6. Anant Agarwal and Jeffrey Lang: Foundations of Analog and Digital Electronic Circuits, Morgan Kaufmann Series, Jul 15, 2005